SATYAMEVA AEGIS

Blockchain-Based Secure Platform for Identity, Access Control, and Digital Asset Management

Smart India Hackathon 2025 · Ministry of Education's Innovation Cell · Technology Bucket: Blockchain & Cybersecurity

◆ 1. Idea Overview

Satyameva Aegis is a production-ready, full-stack blockchain credential verification platform built to eliminate academic document fraud and enable instant digital identity verification across Indian universities. The platform combines a proof-of-work blockchain ledger, AI-assisted document data extraction (Google Gemini), cryptographic PBKDF2-HMAC-SHA256 hashing, institutional QR code branding, and a multi-role web portal (Student · Issuer · Verifier · Admin) — all backed by a scalable Flask + SQLAlchemy backend.

Developed as a live prototype for the Jaypee Group of Institutions (JIIT Noida · JUIT Waknaghat · JUET Guna), the system is immediately deployable and has passed a comprehensive 11-suite E2E automated test battery and full-coverage smoke tests across all routes and credential states.

◆ 2. Problem Statement

The Core Challenge: India produces over 38 million students annually. Academic document fraud — forged marksheets, fake degrees, and tampered certificates — costs employers, institutions, and the government billions in verification overhead and fraudulent hires. Traditional paper-based credentials are trivially forged, and existing digital verification portals are institution-siloed, slow, and unauthenticated.

Specific gaps this platform addresses:

- No tamper-proof, immutable record trail for issued academic documents.
- Verification requires manual email/call to the issuing registrar — taking days.
- No real-time public credential verification link usable by employers in under 5 seconds.
- No multi-campus institutional QR code standard with cryptographic anchoring.
- No AI-assisted batch issuance pipeline for registrar offices handling thousands of credentials.

◆ 3. Proposed Solution

Satyameva Aegis solves this with a tri-layer architecture:

Layer	Component	Technology	Function
1 — Trust	Blockchain Ledger	Proof-of-Work (SHA-256, Difficulty 4)	Immutable credential anchoring — any tampering breaks the chain.
2 — Identity	Cryptographic Hashing	PBKDF2-HMAC-SHA256, 100,000 iter.	Per-doc salted hash; re-derived on every verify — no silent pass for edited rows.
3 — Access	Role-Based Access Control	Flask Blueprints + JWT-like session	Student / Issuer / Verifier / Admin isolation with ownership checks.
4 — Intelligence	AI Extraction	Google Gemini 2.0 Flash	Auto-extracts credential metadata from uploaded PDFs/images for bulk issuance.
5 — Verification	QR + Public Link	HMAC-signed URL + Error-Correction H QR	Permanent, campus-branded verification QR — lifetime valid, cryptographically anchored.

◆ 4. Key Features

■ **Blockchain Credential Anchoring** — Every issued credential is hashed and mined into a proof-of-work chain. Employers get a 6-verdict answer (VALID / EXPIRED / REVOKED / TAMPERED / NOT_FOUND / QR_EXPIRED) in under 2 seconds.

■ **Gemini AI Auto-Extraction** — Upload a PDF/image; Gemini extracts structured credential data automatically, eliminating manual entry for bulk issuance.

■ **Bulk CSV Issuance** — All-or-nothing atomic batch validator: register thousands of credentials from a spreadsheet. Any single row error aborts the entire batch — 0 partial writes.

■ **Multi-Campus Institutional QR** — Campus-specific QR codes embed JIIT / JUIT / JUET crests with Error Correction Level H. Permanent, lifetime-valid verification links with no expiration.

■ **Automated Email Notifications** — Students receive instant Gmail SMTP notifications on issuance and verification results (hackathonsage@gmail.com).

■■ **Stamped PDF Certificates** — Download verification-grade stamped PDFs with embedded QR, blockchain metadata, and institutional branding.

■ **Admin Control Center** — Master admin manages all institutions, users, documents, and the live blockchain ledger from a dedicated admin portal.

■■ **Native Desktop App Mode** — Silent VBScript launcher opens the platform in frameless Chrome/Edge app window with a custom Cyber Shield desktop icon.

◆ 5. Technology Stack

Component	Technology	Purpose
Backend	Python 3.14 + Flask 3.0	Web framework, blueprint routing, session management
Database	SQLite + Flask-SQLAlchemy	Credential, block, user, verification log storage
Blockchain	Custom PoW (SHA-256)	Immutable tamper-proof credential ledger (blockchain.py)
Cryptography	PBKDF2-HMAC-SHA256	Per-document salted hashing (security.py)
AI Engine	Google Gemini 2.0 Flash	Document data extraction from PDF/image uploads
Auth	Flask-Login + Google OAuth 2.0	Role-based access control, institutional login
QR Engine	qrcode + Pillow (ECL-H)	Branded QR generation with institutional crests
Email	Flask-Mail + Gmail SMTP	Real-time credential and verification notifications
Frontend	HTML5 + CSS3 + Three.js (WebGL)	3D blockchain hero, holographic UI, radar HUD animations
Containerization	Docker + docker-compose	Production deployment configuration

◆ 6. Architecture Overview

The platform follows a 5-layer security architecture:

Layer 5 — Presentation	Holographic 3D Web UI (Three.js WebGL + CSS3 Animations)
Layer 4 — Application	Flask Blueprints: auth · student · issuer · verifier · admin · public
Layer 3 — Intelligence	Gemini AI Extraction · QR Code Engine · Email Notification Service
Layer 2 — Cryptographic	PBKDF2-HMAC-SHA256 Hashing · Per-Document Salting · HMAC-Signed QR Tokens
Layer 1 — Blockchain	Proof-of-Work Ledger (Difficulty 4) · Dual-Persisted (SQLite + blockchain.json)

◆ 7. Scalability & National Impact

Phase 1 (Current Prototype): Deployed for Jaypee Group of Institutions — JIIT (Noida), JUIT (Waknaghat), JUET (Guna). Live Gmail SMTP, Gemini AI, Google OAuth, and permanent QR verification.

Phase 2 (National Scale): Extend to NIT, IIT, and state universities via institution onboarding portal. REST API exposure for government DigiLocker and National Academic Depository (NAD) integration.

Phase 3 (Digital Identity Platform): Expand beyond academic credentials to government identity documents, professional certifications, and digital asset provenance — creating a Universal Indian Digital Trust Ledger.

Estimated Impact:

- Eliminate document fraud for 38M+ Indian students annually.
- Reduce employer credential verification time from days → under 5 seconds.
- Provide registrar offices AI-assisted bulk issuance for 10,000+ credential batch operations.
- Interoperate with DigiLocker / NAD via API bridge for national digital ID ecosystem.

◆ 8. Prototype Status & Testing

The prototype is **fully functional and production-ready** as of SIH 2025 submission date.

Test Suite	Coverage	Result
_smoke.py — Route Smoke Test	All routes: login, dashboard, credential detail, QR, download, public verify	■ ALL PASS
test_e2e.py — E2E Integration	11 full integration suites across all user roles and credential states	■ 11/11 PASS (100%)
_assert.py — Behavioral Test	6-verdict logic, QR freshness/staleness, RBAC blocking, idle timeout	■ ALL PASS
test_bulk_issue_e2e.py	4-tier bulk issuance: CSV parsing, atomic all-or-nothing validation, batch mining	■ ALL PASS

◆ 9. Live Demo & Access

Live Prototype Demo: http://127.0.0.1:5000 (Local deployment — run `python run.py` from project root)

Walkthrough Demo Sequence (5 minutes):

1. Sign in as a student — see the live 3D blockchain rendered in WebGL on the dashboard.
2. Open a credential — view the cryptographic seal pressed onto the certificate, flip to see live blockchain ledger data.
3. Scan the permanent QR code — verify the credential on a public link in under 2 seconds.
4. Open the Issuer portal — use Gemini AI extraction to auto-fill a credential from a PDF upload, then bulk-issue via CSV.
5. Check the Admin panel — view all institutions, users, documents, and the live blockchain block explorer.